

Project 2

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1 Introduction

This report presents a regression and symbolic regression analysis across three benchmark datasets. The analysis includes multiple regression, regularized regression using Ridge and Lasso, forward feature selection, transformed regression, and symbolic regression. ScalaTion (Scala 3) is used across all five modeling approaches, while Python's `statsmodels` is used for multiple and regularized regression and `PySR` is used for symbolic regression. Across the required cases, model results and evaluation measures are presented and discussed to examine model fit, predictive performance, and the effects of different regression techniques.

2 Multiple Regression

Multiple regression is used to predict a single outcome from multiple independent variables while estimating the association between each predictor and the response conditional on the other predictors. For each dataset, a multiple regression model was fitted using all available predictors in both ScalaTion and `statsmodels`, giving six required regression models. Additional simple regression models were fitted using the two predictors with the largest absolute correlations with the response variable. The same predictor selections were used for the ScalaTion and `statsmodels` analyses. All R^2 values reported for the full regression models are in-sample values computed using the full datasets and are therefore not directly comparable to held-out test-set metrics reported in later sections.

2.1 ScalaTion

The ScalaTion implementation used an intercept term and all available predictors for each of the multiple regression models. The fitted models were evaluated using the reported R^2 , adjusted R^2 , residual standard error, and fitted coefficient statistics. Simple regression models were also fitted for the two predictors with the largest absolute correlations with each response variable.

2.1.1 Auto MPG

For the Auto MPG dataset, the full multiple regression model uses cylinders, displacement, horsepower, weight, acceleration, model year, and origin to predict MPG. The resulting model showed a strong overall fit, with an R^2 of approximately 0.821. Substantial multicollinearity was present among several vehicle characteristics. The VIF values from ScalaTion were above 10 for cylinders, displacement, and weight, with displacement having the highest VIF at approximately 21.84. The two predictors with the largest absolute correlations with MPG were weight and displacement. Both simple regression models showed a negative relationship with MPG, with higher vehicle weight and displacement corresponding to lower predicted MPG.

Table 1: Auto MPG full regression coefficients from ScalaTion.

Predictor	Estimate	Std. Error	t	p	VIF
Intercept	-17.2184	4.6443	-3.71	0.0002	—
Cylinders	-0.4934	0.3233	-1.53	0.1270	10.74
Displacement	0.01990	0.00752	2.65	0.0081	21.84
Horsepower	-0.01695	0.01379	-1.23	0.2189	9.94
Weight	-0.00647	0.00065	-9.93	<0.001	10.83
Acceleration	0.0806	0.0988	0.82	0.4150	2.63
Model year	0.7508	0.0510	14.73	<0.001	1.24
Origin	1.4261	0.2781	5.13	<0.001	1.77

For Auto MPG, weight, model year, and origin were statistically significant predictors in the full model at $p < 0.001$. Weight had a negative coefficient, while model year and origin had positive coefficients. Displacement also had a positive coefficient and was significant at $p = 0.008$, despite its negative simple-regression relationship with MPG. This difference illustrates the effect of controlling for correlated predictors in a multiple regression model. The high VIF values for displacement, cylinders, and weight indicate substantial multicollinearity, so individual coefficient estimates should be interpreted with caution.

```

=====
OLS Regression Results
=====
Dep. Variable:      mpg      R-squared:      0.693
Model:              OLS      Adj. R-squared:    0.692
Method:              Least Squares      F-statistic:    878.8
Date:                Mon, 21 Sep 2026      Prob (F-statistic): 6.02e-102
Time:                09:07:01      Log-Likelihood:  -1130.0
No. Observations:    392      AIC:              2264.
Df Residuals:        390      BIC:              2272.
Df Model:             1
Covariance Type:     nonrobust
=====
              coef      std err      t      P>|t|      [0.025      0.975]
-----
const         46.2165      0.799      57.867      0.000      44.646      47.787
weight        -0.0076      0.000     -29.645      0.000     -0.008     -0.007
=====
Omnibus:         41.682      Durbin-Watson:    0.808
Prob(Omnibus):    0.000      Jarque-Bera (JB):  60.039
Skew:             0.727      Prob(JB):         9.18e-14
Kurtosis:         4.251      Cond. No.         1.13e+04
=====

```

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 1.13e+04. This might indicate that there are strong multicollinearity or other numerical problems.

Figure 1: Auto MPG simple regression using vehicle weight.

```

=====
OLS Regression Results
=====
Dep. Variable:      mpg      R-squared:      0.648
Model:              OLS      Adj. R-squared:    0.647
Method:              Least Squares      F-statistic:    718.7
Date:                Mon, 21 Sep 2026      Prob (F-statistic): 1.66e-90
Time:                09:07:01      Log-Likelihood: -1156.4
No. Observations:    392      AIC:              2317.
Df Residuals:        390      BIC:              2325.
Df Model:             1
Covariance Type:     nonrobust
=====
              coef      std err      t      P>|t|      [0.025      0.975]
-----
const         35.1206      0.494      71.033      0.000      34.149      36.093
displacement   -0.0601      0.002     -26.808      0.000     -0.064     -0.056
=====
Omnibus:         41.308      Durbin-Watson:    0.926
Prob(Omnibus):    0.000      Jarque-Bera (JB):  61.139
Skew:             0.709      Prob(JB):         5.30e-14
Kurtosis:         4.317      Cond. No.         466.
=====

```

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Figure 2: Auto MPG simple regression using vehicle displacement.

2.1.2 Concrete

For the Concrete dataset, the full multiple regression model used cement, blast furnace slag, fly ash, water, superplasticizer, coarse aggregate, fine aggregate, and age to predict compressive strength. The resulting model showed an R^2 of approximately 0.615. The two predictors with the largest absolute correlations with compressive strength were cement and superplasticizer, and both showed positive simple regression relationships with compressive strength. The Concrete predictors also showed some multicollinearity. Several VIF values were above 5, including cement, blast furnace slag, fly ash, water, coarse aggregate, and fine aggregate. These elevated VIF values indicate that several predictors are correlated with the other predictors in the model, which can make individual coefficient estimates more difficult to interpret.

Table 2: Concrete full regression coefficients from ScalaTion.

Predictor	Estimate	Std. Error	t	p	VIF
Intercept	-23.1638	26.5884	-0.87	0.3836	—
Cement	0.1198	0.0085	14.11	<0.001	7.49
Blast furnace slag	0.1038	0.0101	10.25	<0.001	7.28
Fly ash	0.0879	0.0126	6.99	<0.001	6.17
Water	-0.1503	0.0402	-3.74	0.0002	7.00
Superplasticizer	0.2907	0.0935	3.11	0.0019	2.97
Coarse aggregate	0.0180	0.0094	1.92	0.0549	5.08
Fine aggregate	0.0202	0.0107	1.88	0.0597	7.01
Age	0.1142	0.0054	21.05	<0.001	1.12

For Concrete, cement, blast furnace slag, fly ash, water, superplasticizer, and age were statistically significant predictors at the 0.05 level. Water had a negative coefficient, while the other significant predictors had positive coefficients. Coarse aggregate and fine aggregate were not significant at the 0.05 level, with p-values of approximately 0.055 and 0.060, respectively. The elevated VIF values for several material variables indicate substantial correlation among some of the predictors.

OLS Regression Results						
Dep. Variable:	compressive_strength_mpa	R-squared:				0.248
Model:	OLS	Adj. R-squared:				0.247
Method:	Least Squares	F-statistic:				338.7
Date:	Mon, 21 Sep 2026	Prob (F-statistic):				1.32e-65
Time:	09:07:02	Log-Likelihood:				-4214.6
No. Observations:	1030	AIC:				8433.
Df Residuals:	1028	BIC:				8443.
Df Model:	1					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
const	13.4428	1.297	10.365	0.000	10.898	15.988
cement	0.0796	0.004	18.405	0.000	0.071	0.088
Omnibus:	19.695		Durbin-Watson:			1.012
Prob(Omnibus):	0.000		Jarque-Bera (JB):			17.890
Skew:	0.271		Prob(JB):			0.000130
Kurtosis:	2.649		Cond. No.			861.

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Figure 3: Concrete simple regression using cement.

OLS Regression Results						
Dep. Variable:	compressive_strength_mpa	R-squared:				0.134
Model:	OLS	Adj. R-squared:				0.133
Method:	Least Squares	F-statistic:				159.1
Date:	Mon, 21 Sep 2026	Prob (F-statistic):				5.08e-34
Time:	09:07:02	Log-Likelihood:				-4287.1
No. Observations:	1030	AIC:				8578.
Df Residuals:	1028	BIC:				8588.
Df Model:	1					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
const	29.4668	0.699	42.165	0.000	28.095	30.838
superplasticizer	1.0239	0.081	12.614	0.000	0.865	1.183
Omnibus:	24.083		Durbin-Watson:			1.052
Prob(Omnibus):	0.000		Jarque-Bera (JB):			24.387
Skew:	0.353		Prob(JB):			5.06e-06
Kurtosis:	2.738		Cond. No.			12.5

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Figure 4: Concrete simple regression using superplasticizer.

2.1.3 Airfoil

For the Airfoil dataset, the full multiple regression model used frequency, angle of attack, chord length, free-stream velocity, and suction displacement thickness to predict scaled sound pressure. The full model produced an R^2 of approximately 0.516. The two predictors with the largest absolute correlations with scaled sound pressure were frequency and suction displacement thickness. Both simple regression models showed a negative relationship with the response. Compared with the other two datasets, the Airfoil predictors showed lower multicollinearity, with all VIF values below 3.5. Angle of attack had the highest VIF at approximately 3.44, while the remaining predictors had VIF values below 2.6.

Table 3: Airfoil full regression coefficients from ScalaTion.

Predictor	Estimate	Std. Error	t	p	VIF
Intercept	132.8338	0.5447	243.87	<0.001	—
Frequency	-0.00128	0.00004	-30.45	<0.001	1.14
Angle of attack	-0.4219	0.0389	-10.85	<0.001	3.44
Chord length	-35.6880	1.6304	-21.89	<0.001	1.51
Free-stream velocity	0.0999	0.0081	12.28	<0.001	1.04
Suction displacement thickness	-147.3005	15.0147	-9.81	<0.001	2.53

For Airfoil, all five predictors were statistically significant at $p < 0.001$. Frequency, angle of attack, chord length, and suction displacement thickness had negative coefficients, while free-stream velocity had a positive coefficient. Despite the statistical significance of the predictors, the model explained approximately 51.6% of the variation in scaled sound pressure. The relatively modest VIF values suggest that multicollinearity was less severe than in the Auto MPG and Concrete models.

```

=====
OLS Regression Results
=====
Dep. Variable:    scaled_sound_pressure_db    R-squared:        0.153
Model:            OLS                        Adj. R-squared:   0.152
Method:            Least Squares             F-statistic:      270.4
Date:              Mon, 21 Sep 2026           Prob (F-statistic): 5.36e-56
Time:              09:07:02                   Log-Likelihood:   -4910.5
No. Observations:  1503                      AIC:              9825.
Df Residuals:      1501                      BIC:              9836.
Df Model:           1
Covariance Type:   nonrobust
=====
                    coef    std err          t      P>|t|      [0.025    0.975]
-----
const             127.3837      0.222     572.944      0.000      126.868      127.740
frequency_hz      -0.0009      5.2e-05    -16.444      0.000      -0.001      -0.001
=====
Omnibus:            4.973    Durbin-Watson:      0.255
Prob(Omnibus):      0.083    Jarque-Bera (JB):    5.002
Skew:               -0.126    Prob(JB):             0.0020
Kurtosis:           2.073    Cond. No.             5.80e+03
=====

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 5.8e+03. This might indicate that there are
strong multicollinearity or other numerical problems.

```

Figure 5: Airfoil simple regression using frequency.

```

=====
OLS Regression Results
=====
Dep. Variable:    scaled_sound_pressure_db    R-squared:        0.098
Model:            OLS                        Adj. R-squared:   0.097
Method:            Least Squares             F-statistic:      162.6
Date:              Mon, 21 Sep 2026           Prob (F-statistic): 1.92e-35
Time:              09:07:02                   Log-Likelihood:   -4957.6
No. Observations:  1503                      AIC:              9919.
Df Residuals:      1501                      BIC:              9930.
Df Model:           1
Covariance Type:   nonrobust
=====
                    coef    std err          t      P>|t|      [0.025    0.975]
-----
const             126.6632      0.222     571.527      0.000      126.228      127.098
suction_displacement_thickness_m -164.0275      12.862    -12.753      0.000     -189.256     -138.798
=====
Omnibus:            26.070    Durbin-Watson:      0.364
Prob(Omnibus):      0.000    Jarque-Bera (JB):    25.468
Skew:               -0.280    Prob(JB):             2.95e-06
Kurtosis:           2.694    Cond. No.             76.1
=====

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

```

Figure 6: Airfoil simple regression using suction displacement thickness.

2.2 statsmodels

The same three datasets and predictor selections were analyzed using Python’s `statsmodels` ordinary least squares (OLS) implementation. The full models used all available predictors, while the simple models used the two predictors with the largest absolute response correlations. The full Auto MPG model achieved an R^2 of 0.8215 and an adjusted R^2 of 0.8182, with a residual standard error of approximately 3.33 MPG. The Concrete model achieved an R^2 of 0.6155 and an adjusted R^2 of 0.6125, with a residual standard error of approximately 10.40 MPa. The Airfoil model achieved an R^2 of 0.5157 and an adjusted R^2 of 0.5141, with a residual standard error of approximately 4.81 dB.

Table 4: Full multiple regression evaluation results from **statsmodels**.

Dataset	n	R^2	Adjusted R^2	Residual SE	F
Auto MPG	392	0.8215	0.8182	3.3277	252.43
Concrete	1030	0.6155	0.6125	10.3998	204.27
Airfoil	1503	0.5157	0.5141	4.8089	318.82

Table 5: Auto MPG full regression coefficients from **statsmodels**.

Predictor	Estimate	Std. Error	t	p
Intercept	-17.2184	4.644	-3.707	<0.001
Cylinders	-0.4934	0.323	-1.526	0.128
Displacement	0.0199	0.008	2.647	0.008
Horsepower	-0.0170	0.014	-1.230	0.220
Weight	-0.0065	0.001	-9.929	<0.001
Acceleration	0.0806	0.099	0.815	0.415
Model year	0.7508	0.051	14.729	<0.001
Origin	1.4261	0.278	5.127	<0.001

Table 6: Concrete full regression coefficients from **statsmodels**.

Predictor	Estimate	Std. Error	t	p
Intercept	-23.1638	26.588	-0.871	0.384
Cement	0.1198	0.008	14.110	<0.001
Blast furnace slag	0.1038	0.010	10.245	<0.001
Fly ash	0.0879	0.013	6.988	<0.001
Water	-0.1503	0.040	-3.741	<0.001
Superplasticizer	0.2907	0.093	3.110	0.002
Coarse aggregate	0.0180	0.009	1.919	0.055
Fine aggregate	0.0202	0.011	1.883	0.060
Age	0.1142	0.005	21.046	<0.001

Table 7: Airfoil full regression coefficients from **statsmodels**.

Predictor	Estimate	Std. Error	t	p
Intercept	132.8338	0.545	243.866	<0.001
Frequency	-0.0013	0.00004	-30.452	<0.001
Angle of attack	-0.4219	0.039	-10.847	<0.001
Chord length	-35.6880	1.630	-21.889	<0.001
Free-stream velocity	0.0999	0.008	12.279	<0.001
Suction displacement thickness	-147.3005	15.015	-9.810	<0.001

The **statsmodels** simple regression results were consistent with the correlation analysis. For Auto MPG, weight produced an R^2 of 0.6926, while displacement produced an R^2 of 0.6482. For Concrete, cement produced an R^2 of 0.2478, while superplasticizer produced an R^2 of 0.1340. For Airfoil, frequency produced an R^2 of 0.1527, while suction displacement thickness produced an R^2 of 0.0978. All six simple regression models had statistically significant predictor coefficients with $p < 0.001$.

2.3 Results Comparison

The ScalaTion and **statsmodels** implementations produced consistent results across all three datasets. Both implementations produced the same full-model R^2 , adjusted R^2 , residual standard error, and F-statistic to the displayed precision. The agreement between the two implementations is expected because both fit the same ordinary least squares multiple regression model using the same data and predictors.

For Auto MPG, the full model explained approximately 82.1% of the variation in MPG. Weight had the largest absolute correlation with MPG among the predictors and produced a stronger simple regression fit than displacement. For Concrete, the full model explained approximately 61.5% of the variation in compressive strength, while the individual simple regression models explained substantially less variation. For Airfoil, the full model explained approximately 51.6% of the variation in scaled sound pressure, while the individual simple regression models explained less than 16%.

The results also demonstrate the importance of considering relationships among predictors rather than evaluating coefficients independently. Auto MPG had particularly strong multicollinearity among cylinders, displacement, horsepower, and weight, while Concrete had moderate multicollinearity among several material composition variables. The Airfoil predictors showed comparatively lower multicollinearity. These differences help explain why the full regression models captured more variation in the response than the individual simple regressions, while multicollinearity made some individual coefficient estimates more difficult to interpret.

3 Regularized Regression

Regularized regression is used to reduce overfitting by adding a penalty to the regression coefficients. For the Auto MPG dataset, Ridge and Lasso regression are applied to examine how coefficient shrinkage affects model performance when predicting MPG. Ridge shrinks predictor coefficients toward zero, while Lasso can additionally reduce coefficients to exactly zero, effectively performing variable selection. Both methods are implemented using `ScalaTion` and `statsmodels`.

3.1 Statsmodels

3.1.1 Ridge Regression

```
Alpha: 0.001, Average RMSE: 3.3532
Alpha: 0.01, Average RMSE: 3.3490
Alpha: 0.1, Average RMSE: 3.4184
Alpha: 1, Average RMSE: 3.8973
Alpha: 10, Average RMSE: 6.1356
Alpha: 100, Average RMSE: 7.6559
```

```
Best Ridge Alpha: 0.01
```

Figure 7: Ridge alpha tuning using 5-fold cross-validation.

The Ridge regularization parameter was tuned using 5-fold cross-validation on the training data. Candidate values of $\alpha \in \{0.001, 0.01, 0.1, 1, 10, 100\}$ were evaluated using average validation RMSE. The value $\alpha = 0.01$ produced the lowest average RMSE and was selected for the final Ridge model.

```

const          23.599361
cylinders      -0.469692
displacement   0.977049
horsepower     -0.908489
weight         -4.667132
acceleration   0.006313
model_year     2.722611
origin         1.268601
dtype: float64

```

Figure 8: Ridge regression coefficients.

The fitted Ridge model retains all predictors while shrinking their coefficients through regularization. Among the standardized predictors, weight has the largest coefficient magnitude at approximately -4.67 , indicating a strong negative association with MPG, while model year has a relatively large positive coefficient of approximately 2.72 . Acceleration has a coefficient close to zero, indicating a relatively small contribution to the fitted model.

	Actual MPG	Predicted MPG
0	26.0	25.890771
1	21.6	26.167406
2	36.1	34.297227
3	26.0	24.800514
4	27.0	28.494060

Figure 9: Actual and predicted MPG values from the Ridge model.

The fitted output compares the actual MPG values from the test data with the MPG values predicted by the Ridge model. The predictions generally follow the observed MPG values, although some prediction error remains.

```

Ridge RMSE: 3.2973096237408797
Ridge R2: 0.7869881168144173

```

Figure 10: Ridge regression evaluation metrics.

Ridge regression achieved an R^2 of 0.787, meaning the model explains approximately 78.7% of the variation in MPG in the test data. The RMSE of 3.297 indicates that the model's MPG predictions have a typical error magnitude of about 3.3 MPG.

Overall, Ridge regression provides a reasonably strong fit while using regularization to shrink the predictor coefficients and reduce the risk of overfitting. This is particularly useful for the Auto MPG data because several predictors, including weight, displacement, horsepower, and cylinders, are strongly related to one another, as shown in Project 1. Ridge allows the model to retain all of these predictors while shrinking their coefficients rather than eliminating them. Thus, Ridge maintains information from the different vehicle characteristics while controlling the influence of highly related predictors.

3.1.2 Lasso Regression

```
Alpha: 0.001, Average RMSE: 3.3536
Alpha: 0.01, Average RMSE: 3.3437
Alpha: 0.1, Average RMSE: 3.3518
Alpha: 1, Average RMSE: 3.6104
Alpha: 10, Average RMSE: 7.9141
Alpha: 100, Average RMSE: 7.9141

Best Lasso Alpha: 0.01
```

Figure 11: Lasso alpha tuning using 5-fold cross-validation.

The Lasso regularization parameter was tuned using 5-fold cross-validation on the training data. Candidate values of $\alpha \in \{0.001, 0.01, 0.1, 1, 10, 100\}$ were evaluated using average validation RMSE. The value $\alpha = 0.01$ produced the lowest average RMSE and was selected for the final Lasso model.

```
const          23.599361
cylinders      -0.382769
displacement    1.128755
horsepower     -0.839027
weight        -4.975586
acceleration    0.000000
model_year     2.764608
origin         1.258881
dtype: float64
```

Figure 12: Lasso regression coefficients.

The fitted Lasso model shrinks the predictor coefficients through regularization and reduces the coefficient for acceleration to zero, effectively removing it as a predictor. Weight has the largest coefficient magnitude among the remaining standardized predictors, while model year has a relatively large positive coefficient.

	Actual MPG	Predicted MPG
0	26.0	25.837619
1	21.6	26.086561
2	36.1	34.424332
3	26.0	24.822670
4	27.0	28.448806

Figure 13: Actual and predicted MPG values from the Lasso model.

The fitted output compares the actual MPG values from the test data with the MPG values predicted by the Lasso model. The predicted values generally follow the actual MPG values while still showing some prediction error.

Lasso RMSE: 3.2906920560669057
Lasso R^2 : 0.7878422713750637

Figure 14: Lasso regression evaluation metrics.

Lasso regression achieved an R^2 of 0.7878, meaning the model explains approximately 78.8% of the variation in MPG in the test data. The RMSE of 3.291 indicates that the model's MPG predictions have a typical error magnitude of about 3.29 MPG.

Overall, Lasso regression provides a reasonably strong fit while using regularization to shrink coefficients. In this model, Lasso reduced the coefficient for acceleration to zero, effectively removing it as a predictor. This is consistent with the Auto MPG analysis in Project 1, where acceleration had a weaker relationship with MPG than predictors such as weight, displacement, horsepower, and cylinders. Thus, Lasso maintained strong predictive performance while producing a more selective model.

3.2 ScalaTion

3.2.1 Ridge Regression

```
Ridge lambda tuning:
| RidgeRegression with lambda = 0.1 has sse = 713.7525699365118 |
| RidgeRegression with lambda = 0.2 has sse = 713.5365867814872 |
| RidgeRegression with lambda = 0.4 has sse = 713.1495239145677 |
| RidgeRegression with lambda = 0.8 has sse = 712.5362970863046 |
| RidgeRegression with lambda = 1.6 has sse = 711.8303659095648 |
| RidgeRegression with lambda = 3.2 has sse = 711.8153255080763 |
| RidgeRegression with lambda = 6.4 has sse = 714.4740228233744 |
| RidgeRegression with lambda = 12.8 has sse = 722.4991830313115 |
| RidgeRegression with lambda = 25.6 has sse = 737.7183111025892 |
| RidgeRegression with lambda = 51.2 has sse = 762.7016504571261 |
| RidgeRegression with lambda = 102.4 has sse = 807.6552158726756 |
| RidgeRegression with lambda = 204.8 has sse = 897.1513975453229 |
| RidgeRegression with lambda = 409.6 has sse = 1076.40580565289 |
| RidgeRegression with lambda = 819.2 has sse = 1409.7306277835444 |
| RidgeRegression with lambda = 1638.4 has sse = 1919.7235723580118 |
| RidgeRegression with lambda = 3276.8 has sse = 2505.485491122153 |
| RidgeRegression with lambda = 6553.6 has sse = 3010.629751886936 |
| RidgeRegression with lambda = 13107.2 has sse = 3359.4603911741638 |
| RidgeRegression with lambda = 26214.4 has sse = 3567.8769296775936 |
| RidgeRegression with lambda = 52428.8 has sse = 3682.349877269195 |

Best Ridge Lambda: 3.2
Best Ridge CV SSE: 711.8153
```

Figure 15: ScalaTion Ridge lambda tuning using cross-validation.

The ScalaTion Ridge regularization parameter was tuned using cross-validation on the training data. Candidate lambda values were evaluated using cross-validated SSE, where lower SSE indicates better validation performance. The lowest CV SSE of 711.815 was obtained at $\lambda = 3.2$, which was selected for the final Ridge model.

```
Ridge coefficients:
Intercept = 23.416561
cylinders      -1.078994
displacement   1.232053
horsepower    -0.542461
weight        -4.738602
acceleration   0.007586
model_year     2.651239
origin         1.139237

Ridge actual vs predicted MPG:
Actual: 34.300 Predicted: 31.613
Actual: 19.000 Predicted: 26.525
Actual: 15.000 Predicted: 15.493
Actual: 33.700 Predicted: 33.805
Actual: 29.800 Predicted: 33.607

Ridge RMSE: 3.556045
Ridge R²: 0.779972
```

Figure 16: ScalaTion Ridge regression coefficients, fitted output, and evaluation metrics.

The ScalaTion Ridge model achieved an R^2 of 0.7800, meaning the model explains approximately 78.0% of the variation in MPG in the test data. The RMSE of 3.556 indicates that the model's MPG predictions have a typical error magnitude of about 3.56 MPG. The fitted output also shows that the predicted MPG values generally follow the actual test values, although some prediction error remains.

Overall, the ScalaTion Ridge model provides a reasonably strong fit while retaining all predictors and shrinking their coefficients through regularization. Weight has the largest coefficient magnitude among the standardized predictors at approximately -4.739 , indicating a strong negative association with MPG in the fitted model. Model year has a positive coefficient of approximately 2.651, while acceleration has a coefficient close to zero. As with the Statsmodels Ridge model, Ridge retains these predictors rather than eliminating them.

3.2.2 Lasso Regression

```
Lasso lambda tuning:
| LassoRegression with lambda = 3.2 has sse = 716.4706211833529 |
| LassoRegression with lambda = 6.4 has sse = 715.9913823683123 |
| LassoRegression with lambda = 12.8 has sse = 716.2214580015408 |
| LassoRegression with lambda = 25.6 has sse = 715.414402699883 |
| LassoRegression with lambda = 51.2 has sse = 718.6457373741941 |
| LassoRegression with lambda = 102.4 has sse = 739.1994583170452 |
| LassoRegression with lambda = 204.8 has sse = 827.8038274144692 |
| LassoRegression with lambda = 409.6 has sse = 1147.4621449313456 |
| LassoRegression with lambda = 819.2 has sse = 2378.1516869310367 |
| LassoRegression with lambda = 1638.4 has sse = 6140.51301731296 |
| LassoRegression with lambda = 3276.8 has sse = 14132.35192204406 |
| LassoRegression with lambda = 6553.6 has sse = 37826.29469091059 |
| LassoRegression with lambda = 13107.2 has sse = 37826.29469091059 |
| LassoRegression with lambda = 26214.4 has sse = 37826.29469091059 |
| LassoRegression with lambda = 52428.8 has sse = 37826.29469091059 |
| LassoRegression with lambda = 104857.6 has sse = 37826.29469091059 |
| LassoRegression with lambda = 209715.2 has sse = 37826.29469091059 |
| LassoRegression with lambda = 419430.4 has sse = 37826.29469091059 |
| LassoRegression with lambda = 838860.8 has sse = 37826.29469091059 |
| LassoRegression with lambda = 1677721.6 has sse = 37826.29469091059 |

Best Lasso Lambda: 25.6
Best Lasso CV SSE: 715.4144
```

Figure 17: ScalaTion Lasso lambda tuning using cross-validation.

The ScalaTion Lasso regularization parameter was tuned using cross-validation on the training data. Candidate lambda values were evaluated using cross-validated SSE, where lower SSE indicates better validation performance. The lowest CV SSE of 715.414 was obtained at $\lambda = 25.6$, which was selected for the final Lasso model.

```
Lasso coefficients:
const          23.335032
cylinders      -0.250666
displacement   -0.143925
horsepower     -0.121958
weight        -4.643400
acceleration    0.050525
model_year     2.601568
origin         0.943053

Lasso actual vs predicted MPG:
Actual: 34.300 Predicted: 31.550
Actual: 19.000 Predicted: 26.369
Actual: 15.000 Predicted: 15.727
Actual: 33.700 Predicted: 33.301
Actual: 29.800 Predicted: 33.434

Lasso RMSE: 3.544011
Lasso R2: 0.781459
```

Figure 18: ScalaTion Lasso regression coefficients, fitted output, and evaluation metrics.

The ScalaTion Lasso model achieved an R^2 of 0.7815, meaning the model explains approximately 78.1% of the variation in MPG in the test data. The RMSE of 3.544 indicates that the model's MPG predictions have a typical error magnitude of about 3.54 MPG. The fitted output similarly shows that the predicted MPG values generally follow the actual test values.

Overall, the ScalaTion Lasso model provides a reasonably strong fit while shrinking the predictor coefficients through regularization. Weight has the largest coefficient magnitude among the standardized predictors at approximately -4.643 , while model year has a positive coefficient of approximately 2.602 . Unlike the Statsmodels Lasso model, the ScalaTion Lasso model did not reduce acceleration exactly to zero. Its coefficient remained small at approximately 0.051 . The ScalaTion Ridge and Lasso models therefore produced very similar predictive performance, although the two regularization methods produced different coefficient estimates.

3.3 Summary

Overall, `statsmodels` and ScalaTion produced similar Ridge and Lasso results for the Auto MPG dataset. Ridge achieved an RMSE of 3.297 and R^2 of 0.7870 in `statsmodels`, compared with an RMSE of 3.556 and R^2 of 0.7800 in ScalaTion. Lasso achieved an RMSE of 3.291 and R^2 of 0.7878 in `statsmodels`, compared with an RMSE of 3.544 and R^2 of 0.7815 in ScalaTion. These test metrics should not be read as a comparison between the two tools, however: the `statsmodels` models were evaluated on a split produced by scikit-learn's `train_test_split`, while the ScalaTion models used a different split produced by Scala's random shuffle. The two test sets contain different cars, and the split alone changes the result noticeably; unregularized regression scores a test RMSE of 3.273 on the `statsmodels` split but 3.528 on the ScalaTion split.

The two implementations also used different regularization parameters and tuning procedures. In `statsmodels`, the penalty strength is controlled by α , and was selected using 5-fold cross-validation with average validation RMSE. In ScalaTion, the penalty strength is controlled by λ , and the built-in tuning procedure selects λ using cross-validated SSE. Because the implementations and tuning criteria differ, the numerical values of α and λ should not be directly compared.

Ridge behaved similarly across both implementations by shrinking coefficients while retaining all predictors. Lasso showed a larger difference: the `statsmodels` model reduced acceleration to exactly zero, while ScalaTion retained acceleration with a small coefficient of approximately 0.051 . Within both implementations, Ridge and Lasso had very similar predictive performance, with Lasso having a slightly lower RMSE and slightly higher R^2 . Overall, the results show that both methods provide similar predictive performance, while Lasso can produce a more selective model depending on the implementation and selected regularization strength.

4 Forward Feature Selection

Forward feature selection was run in ScalaTion on two datasets, Auto MPG ($n = 392$) and Concrete ($n = 1,030$). For each dataset, the rows were shuffled once with a fixed random seed (42), and a multiple regression model with an intercept column was built over the available predictors. ScalaTion's `forwardSelAll` routine then started from an intercept-only model and added one variable at a time, at each step choosing the addition that maximized adjusted R^2 (`QoF.rSqBar`). Each candidate model was fitted on ScalaTion's default 80% training split (314 rows for Auto MPG and 824 for Concrete), so the path R^2 and adjusted R^2 values are training-set measures; adjusted R^2 was recomputed with the training-set size, because `forwardSelAll` reports it using the full dataset size. Every candidate model was also evaluated with the default 5-fold cross-validation, so R^2_{cv} is the average out-of-sample R^2 across the five folds. The best model is the step where adjusted R^2 is maximized over the full selection path, and that subset of variables was refit with `trainNtest()` on the full dataset.

4.1 ScalaTion

4.1.1 Auto MPG

Forward selection considered the seven predictors (cylinders, displacement, horsepower, weight, acceleration, model year, and origin) for MPG. Table 8 lists the variable added at each step along with the metrics reported by `forwardSelAll` (all values in %), and Figure 19 plots them against the number of predictors.

Table 8: Forward selection path for Auto MPG (values in %). Adjusted R^2 peaks at step 6. Adjusted R^2 is computed with the training-set size used by `forwardSelAll` (314 rows).

Step	Variable added	R^2	Adjusted R^2	sMAPE	R^2_{cv}
0	intercept	-0.000	0.00	28.410	-1.283
1	weight	70.660	70.57	13.919	68.892
2	model year	81.461	81.34	11.867	80.384
3	origin	82.532	82.36	11.187	80.984
4	cylinders	82.583	82.36	11.113	80.784
5	displacement	82.779	82.50	11.068	80.851
6	horsepower	82.856	82.52	11.173	81.063
7	acceleration	82.866	82.47	11.187	80.855

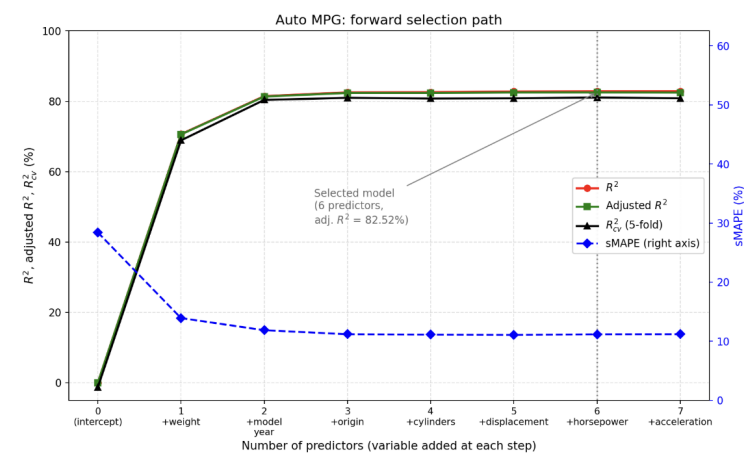


Figure 19: Auto MPG forward selection: R^2 (red), adjusted R^2 (green), sMAPE (blue), and R^2_{cv} (black) versus the number of predictors.

Adjusted R^2 rises sharply with the addition of weight and model year, then flattens. Origin, cylinders, displacement, and horsepower each add only a fraction of a percentage point, and adding acceleration at step 7 decreases adjusted R^2 (from 82.52 to 82.47) even though raw R^2 still increases slightly. This is the expected behavior of adjusted R^2 , which penalizes predictors that do not sufficiently improve the fit, and it signals that acceleration is not worth including. sMAPE is also lowest at step 5 rather than improving continuously, which supports the conclusion that the search has saturated.

The maximum adjusted R^2 occurs at step 6, so the best model keeps six predictors (weight, model year, origin, cylinders, displacement, and horsepower) plus the intercept and drops only acceleration. Refitting this subset on all 392 observations with `trainNtest()` gives the coefficients in Table 9 and the actual versus predicted plot in Figure 20.

Table 9: Coefficients of the best Auto MPG forward selection model (full-data refit).

Variable	Coefficient
intercept	-15.563492
weight	-0.006218
model year	0.747516
origin	1.428242
cylinders	-0.506685
displacement	0.019269
horsepower	-0.023895

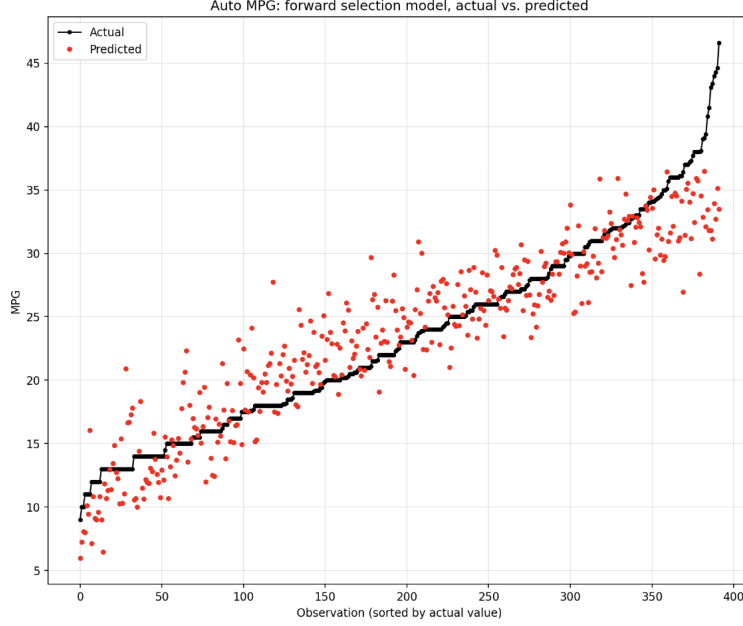


Figure 20: Actual (black) and predicted (red) MPG for the best Auto MPG forward selection model, with observations sorted by actual value.

The selected model achieved an R^2 of 0.8212, an adjusted R^2 of 0.8184, and an RMSE of 3.296 MPG on the full dataset. Its 5-fold cross-validated R^2_{cv} was 0.8106, which is the more honest estimate of out-of-sample performance. This is a large improvement over the best simple regression (weight alone, $R^2 = 0.6926$), and it is essentially the same fit as the full seven-predictor model from the multiple regression section (adjusted R^2 of 0.8182), so dropping acceleration costs nothing. Heavier cars and cars with more cylinders are associated with lower MPG, while later model years are associated with higher MPG, which is consistent with improvements in fuel efficiency over the 1970–1982 sample period. The displacement coefficient is slightly positive even though displacement is negatively correlated with MPG on its own. This is likely because weight, cylinders, and displacement are strongly correlated with one another, so once weight and cylinders are in the model, displacement no longer plays its marginal role.

4.1.2 Concrete

Forward selection considered all eight predictors (cement, blast furnace slag, fly ash, water, superplasticizer, coarse aggregate, fine aggregate, and age) for compressive strength. Table 10 lists the selection path, and Figure 21 plots the metrics against the number of predictors.

Table 10: Forward selection path for Concrete (values in %). Adjusted R^2 peaks at step 8. Adjusted R^2 is computed with the training-set size used by `forwardSelAll` (824 rows).

Step	Variable added	R^2	Adjusted R^2	sMAPE	R^2_{cv}
0	intercept	−0.000	0.00	40.080	−1.465
1	cement	24.566	24.47	36.919	22.857
2	superplasticizer	35.429	35.27	34.587	33.061
3	age	48.513	48.32	30.110	46.730
4	blast furnace slag	54.989	54.77	28.171	53.311
5	water	58.501	58.25	27.518	56.647
6	fly ash	61.754	61.47	26.216	59.805
7	fine aggregate	61.785	61.46	26.162	59.789
8	coarse aggregate	61.888	61.51	26.070	59.749

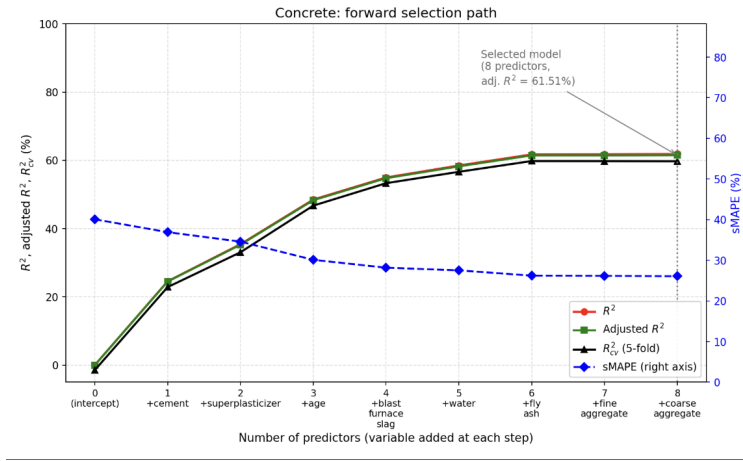


Figure 21: Concrete forward selection: R^2 (red), adjusted R^2 (green), sMAPE (blue), and R^2_{cv} (black) versus the number of predictors.

Unlike Auto MPG, adjusted R^2 for Concrete keeps increasing through nearly the whole path. The one exception is a tiny dip at step 7 (61.47 to 61.46), and step 8 then reaches the maximum, so the full model narrowly wins. The largest gains come from cement and superplasticizer, the two predictors most correlated with compressive strength, followed by a large jump from age. Blast furnace slag, water, and fly ash then add roughly 6.5, 3.5, and 3.2 points of adjusted R^2 respectively, while fine and coarse aggregate together add less than 0.1 points, which appears as the flat tail of the curve in Figure 21. sMAPE drops from 40.1% for the intercept-only model to about 26% by step 6 and barely changes afterward.

Because adjusted R^2 is highest at step 8, the best model keeps all eight predictors plus the intercept. Table 11 gives the coefficients from the full-data refit, and Figure 22 shows the actual versus predicted values.

Table 11: Coefficients of the best Concrete forward selection model (full-data refit).

Variable	Coefficient
intercept	−23.163756
cement	0.119785
superplasticizer	0.290687
age	0.114226
blast furnace slag	0.103847
water	−0.150298
fly ash	0.087943
fine aggregate	0.020154
coarse aggregate	0.018030

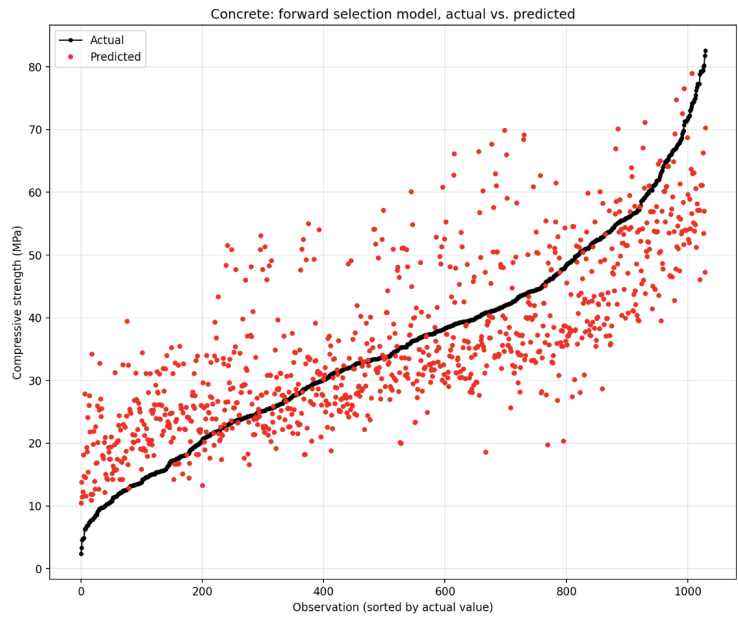


Figure 22: Actual (black) and predicted (red) compressive strength for the best Concrete forward selection model, with observations sorted by actual value.

The selected model achieved an R^2 of 0.6155, an adjusted R^2 of 0.6125, and an RMSE of 10.354 MPa on the full dataset, with a cross-validated R^2_{cv} of 0.5975. This is a marked improvement over the best simple regression (cement alone, $R^2 = 0.2478$) and matches the full multiple regression model. All coefficients are positive except water, which agrees with domain expectations: a higher water content at fixed cement, meaning a higher water-to-cement ratio, is associated with lower strength. The actual versus predicted plot shows much wider scatter than for Auto MPG, particularly for mid-range strengths, which is consistent with the moderate R^2 and suggests that a large share of the variability is not captured by linear terms alone.

4.2 Summary

In both datasets, forward selection guided by adjusted R^2 chose the same leading predictors as the correlation analysis in the multiple regression section: weight came first for Auto MPG, and cement and superplasticizer came first for Concrete. The two datasets behave differently after that. For Auto MPG, adjusted R^2 saturates quickly and declines when acceleration is forced in, so selection stops at six of seven predictors. For Concrete, adjusted R^2 keeps climbing, however modestly, to the full model, so no variables are excluded. In both cases the selected model performs about as well as the full multiple regression model, and the cross-validated R^2_{cv} (0.8106 and 0.5975) is a little lower than the full-data R^2 , as expected. This highlights why adjusted R^2 is a better selection criterion than raw R^2 , which never decreases as predictors are added.

5 Transformed Regression

Transformed regression keeps the linear model and its efficient least-squares training but applies a transformation f to the response:

$$f(y) = \mathbf{b} \cdot \mathbf{x} + \epsilon, \quad \hat{y} = f^{-1}(\mathbf{b} \cdot \mathbf{x}).$$

It was fitted with ScalaTion's `TranRegression` class on two of the three datasets, Auto MPG and Concrete, using all available predictors plus an intercept. For each dataset, five models were compared: plain `Regression` (no transformation) and `TranRegression` with a log, square-root, reciprocal, and Box-Cox transformation,

$$f_\lambda(y) = \frac{y^\lambda - 1}{\lambda} \quad (\lambda \neq 0), \quad f_0(y) = \log y.$$

The transformed variable is the response only: fuel economy (`mpg`, in MPG) for Auto MPG and compressive strength (`compressive_strength_mpa`, in MPa) for Concrete. Predictor variables are left untransformed. Box–Cox was selected as the required alternative to Yeo–Johnson; since every observed response is strictly positive, Yeo–Johnson’s extension to nonpositive values is not needed. The reciprocal transformation is included as an additional comparison.

Because $\lambda = 1$ is a shift of y , the Box–Cox family contains “no transformation” as a special case. The Box–Cox parameter was tuned over $\lambda \in \{-1, -0.75, \dots, 2\}$ by 5-fold cross-validation on the training set only. Predictions were mapped back with f^{-1} , so the comparison tables report R^2 , RMSE, and MAE on the *original* response scale. Full-data fits are descriptive; held-out results come from separate models fitted only to the training rows. Both datasets use an 80/20 split generated by Scala’s random shuffle with seed 42. For Auto MPG, this reproduces the split used by the existing ScalaTion Ridge and Lasso code. Auto MPG uses 314 training and 78 test rows; Concrete uses 824 training and 206 test rows. The already shuffled training rows are divided into five contiguous folds, and CV RMSE pools squared errors across all validation rows. All observed responses are positive. For $\lambda \neq 0$, a valid Box–Cox inverse requires $1 + \lambda z > 0$, even when an integer inverse power would return a finite number outside this domain. Square-root and positive-response reciprocal models require $z \geq 0$ and $z > 0$, respectively. Candidates with any invalid or nonfinite validation prediction are excluded. An invalid full-data or test prediction makes that entire model/split score unavailable; rows are never silently discarded.

Auto MPG and Concrete satisfy the requirement to apply transformed regression to two of the three datasets. They provide a useful comparison because the observed benefit is substantial for Auto MPG and small for Concrete. No transformed-regression result is claimed for Airfoil.

The predictions use direct inverse transformation, without a retransformation bias correction. In general, $f^{-1}(E[f(y) | x])$ is not $E[y | x]$; these fits minimize squared error on the transformed scale, not the original scale. Original-scale validation measures the practical prediction error of this choice. The chosen λ minimizes training CV RMSE; the held-out test set is not used for tuning.

5.1 Auto MPG

Model	In-sample ($n = 392$)			Held-out test		
	R^2	RMSE	MAE	R^2	RMSE	MAE
Regression (no transform)	0.8215	3.294	2.499	0.7835	3.528	2.631
TranRegression, log	0.8562	2.956	2.102	0.8216	3.202	2.285
TranRegression, sqrt	0.8437	3.082	2.252	0.8059	3.340	2.426
TranRegression, reciprocal	0.8490	3.029	2.137	0.8069	3.331	2.274
TranRegression, Box-Cox $\lambda = -0.5$	0.8612	2.904	2.043	0.8279	3.145	2.216

Table 12: Auto MPG: original-scale quality of fit for plain and transformed regression (ScalaTion).

Every transformation improved on plain regression, both in-sample and on the test set. Cross-validation selected $\lambda = -0.5$ (CV RMSE 2.995 versus 3.391 at $\lambda = 1$), and this model reduced test RMSE from 3.528 to 3.145 MPG, a 10.9% reduction. The selected λ lies between the log ($\lambda = 0$) and reciprocal ($\lambda = -1$) transformations. The reciprocal alternative models fuel consumption (gallons per mile, 1/MPG) directly; its improvement over plain regression supports examining nearby powers.

Figure 23 compares full-data fits and residuals. The Box–Cox model improves the fit, but these plots are descriptive and do not replace the held-out evaluation. On the transformed scale, the selected full-data model has $R^2 = 0.890$. Model year ($t = 15.9$) and weight ($t = -10.7$) have the largest absolute t statistics among its predictors. Acceleration has $p = 0.23$. These are conventional OLS statistics conditional on the selected transformation, without adjustment for choosing λ from the data; they are descriptive rather than confirmatory. Coefficients and their standard errors are computed on the transformed scale and are not in MPG units. Origin is treated as a numeric predictor, matching the existing regression setup.

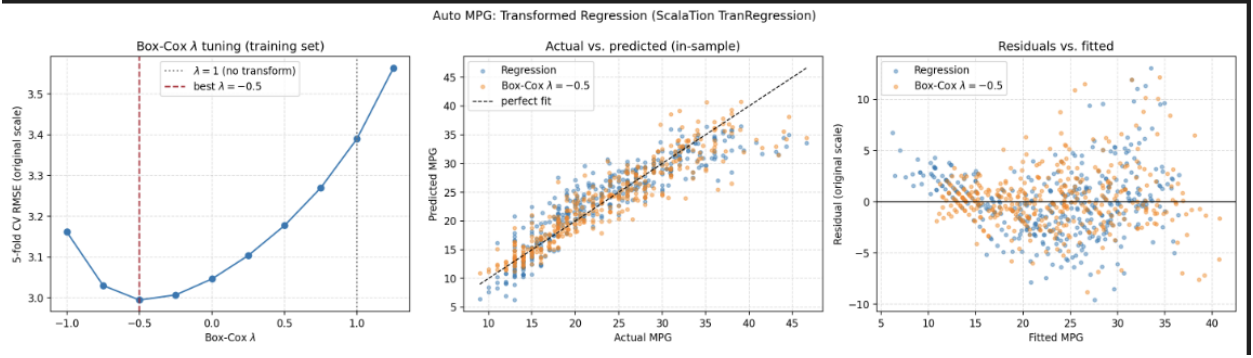


Figure 23: Auto MPG: Box-Cox λ tuning curve (left), actual vs. predicted MPG (center), and residuals vs. fitted values (right) for plain regression and the Box-Cox $\lambda = -0.5$ model.

5.2 Concrete

Model	In-sample ($n = 1030$)			Held-out test		
	R^2	RMSE	MAE	R^2	RMSE	MAE
Regression (no transform)	0.6155	10.354	8.215	0.5957	10.311	8.125
TranRegression, log	0.4559	12.317	9.442	0.4740	11.761	9.305
TranRegression, sqrt	0.5800	10.822	8.637	0.5582	10.779	8.571
TranRegression, reciprocal	—	—	—	—	—	—
TranRegression, Box-Cox $\lambda = 1.25$	0.6218	10.269	8.074	0.6055	10.185	7.970

Table 13: Concrete: original-scale quality of fit. Reciprocal scores are unavailable because the positive-response inverse domain fails for 21 of 1,030 full-data predictions and 1 of 206 test predictions.

Concrete benefits much less from response transformation. Log and square-root transformations increased prediction error relative to plain regression: log reduced test R^2 from 0.596 to 0.474, and square root reduced it to 0.558. The reciprocal transformation failed badly. The model on $1/y$ produces some nonpositive fitted values, outside the inverse domain for positive strengths. Near-zero positive values can also produce very large predictions. This makes the reciprocal model unsuitable here. Cross-validation chose $\lambda = 1.25$, slightly *above* 1, which expands rather than compresses the response. The resulting gain is marginal: test RMSE fell only from 10.311 to 10.185 MPa (1.2%). The valid portion of the CV curve in Figure 24 is nearly flat between $\lambda = 0.75$ and 1.5 (CV RMSE 10.43 at the optimum versus 10.51 with no transformation), so the data provide little evidence for any response transformation.

These results show that transforming only the response offers limited benefit for this linear predictor specification. They do not establish the source of any remaining nonlinearity. Transforming predictors or adding interactions could be investigated separately through symbolic regression. The small improvement on one held-out split is not evidence of a statistically significant gain.

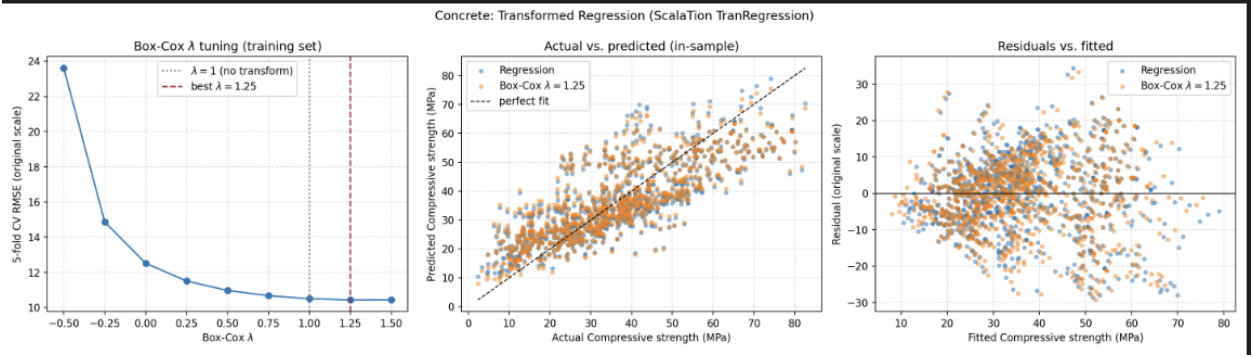


Figure 24: Concrete: Box-Cox λ tuning curve (left; invalid candidates omitted), actual vs. predicted strength (center), and residuals vs. fitted values (right) for plain regression and the Box-Cox $\lambda = 1.25$ model.

5.3 Summary

For Auto MPG, training CV selected Box-Cox $\lambda = -0.5$, and test RMSE fell by 10.9% relative to plain regression. For Concrete, $\lambda = 1.25$ reduced test RMSE by only 1.2%; log and square-root transformations worsened test error, and the reciprocal model produced invalid predictions. These conclusions apply to the tested models and this fixed split, not to all possible ScalaTion or symbolic-regression models. A response transformation should be chosen using validation in the units of the prediction task.

6 Symbolic Regression

Symbolic regression differs from traditional linear regression because it searches not only for model coefficients, but also for the mathematical structure relating the predictors to the response. Instead of assuming a predefined functional form, symbolic regression searches over combinations of mathematical operators and variables to identify equations that balance predictive accuracy and expression complexity.

Two symbolic regression implementations were used in this project: ScalaTion and PySR. ScalaTion constructs transformed candidate features and applies feature selection, while PySR uses an evolutionary search procedure to discover mathematical expressions directly. The three datasets analyzed were Auto MPG, Concrete Compressive Strength, and Airfoil Self-Noise.

6.1 ScalaTion

ScalaTion symbolic regression was applied to all three datasets. Candidate features were constructed from nonlinear transformations of the original predictors, including square-root, squared, inverse, logarithmic, and interaction terms depending on the dataset. Forward feature selection was then used to identify a subset of transformed predictors for the final regression model.

Table 14: ScalaTion symbolic regression performance.

Dataset	R^2	Adjusted R^2	RMSE	MAE
Auto MPG	0.8827	0.8328	2.6101	1.9914
Concrete	0.8180	0.8015	6.9646	5.6196
Airfoil	0.6826	0.6321	3.9647	2.9289

6.1.1 Auto MPG

For Auto MPG, ScalaTion generated nonlinear transformations of vehicle characteristics including weight, model year, displacement, horsepower, acceleration, cylinders, and origin. The final selected model included inverse, square-root, and squared transformations.

Table 15: Selected significant coefficients from the ScalaTion symbolic regression model for Auto MPG.

Term	Estimate	Std. Error	t	p
Intercept	-1016807.1951	517225.1602	-1.97	0.0493
Weight ⁻¹	908334.5295	426892.0794	2.13	0.0334
Model year ²	-10.1170	4.9912	-2.03	0.0427
$\sqrt{\text{model year}}$	83706.6823	42238.6395	1.98	0.0475
Acceleration ⁻¹	808.9958	331.6133	2.44	0.0147
Displacement ⁻²	129732.1150	56783.3200	2.28	0.0223
Weight ⁻²	-626524289.4326	262993286.5486	-2.38	0.0172
Acceleration ⁻²	-3554.8823	1663.9678	-2.14	0.0326
Acceleration ²	0.02863	0.01384	2.07	0.0386
Cylinders ²	1.1834	0.2464	4.80	< 0.001
Cylinders ⁻²	-765.5316	127.7204	-5.99	< 0.001
$\sqrt{\text{cylinders}}$	-114.1268	21.5207	-5.30	< 0.001

The final selected model achieved an R^2 of 0.8827 and an adjusted R^2 of 0.8328. The RMSE was 2.6101 MPG and the MAE was 1.9914 MPG. The residual standard error was approximately 3.1370, and the overall F -statistic was 17.6756.

Several transformed predictors were statistically significant, including inverse weight, squared model year, inverse acceleration, inverse displacement, squared acceleration, and multiple nonlinear transformations of cylinders. These results indicate that the relationship between vehicle characteristics and MPG contains important nonlinear structure.

However, many of the generated transformed terms had extremely large VIF values, with several exceeding millions or even billions. This indicates severe multicollinearity among the symbolic basis functions. Therefore, individual coefficient magnitudes should be interpreted cautiously, and the model is better viewed as a predictive combination of nonlinear transformed features.

6.1.2 Concrete

For the Concrete dataset, ScalaTion generated transformations of cement, slag, fly ash, water, superplasticizer, coarse aggregate, fine aggregate, and age. The selected model contained terms such as $\sqrt{\text{age}}$, cement squared, slag squared, fly ash squared, square-root transformations, and other transformed material variables.

Table 16: Selected significant coefficients from the ScalaTion symbolic regression model for Concrete.

Term	Estimate	Std. Error	t	p
$\sqrt{\text{age}}$	6.6596	0.2074	32.11	< 0.001
Superplasticizer	-1.6063	0.4672	-3.44	< 0.001
Cement ²	0.000105	0.000007	14.14	< 0.001
Age	-0.2441	0.0119	-20.52	< 0.001
Slag ²	0.000142	0.000022	6.30	< 0.001
$\sqrt{\text{water}}$	564.3547	79.3692	7.11	< 0.001
$\sqrt{\text{fine aggregate}}$	-2.1050	0.3517	-5.99	< 0.001
$\sqrt{\text{superplasticizer}}$	6.5834	1.2153	5.42	< 0.001
Water	-32.4357	4.4639	-7.27	< 0.001
Water ²	0.03036	0.00414	7.34	< 0.001

The final selected model achieved an R^2 of 0.8180 and an adjusted R^2 of 0.8015. The RMSE was 6.9646 MPa and the MAE was 5.6196 MPa. The residual standard error was approximately 7.2904, and the overall F -statistic was 49.6891.

For comparison, the full 24-term candidate model produced a slightly higher test R^2 of 0.8378, with an RMSE of 6.4876 and an MAE of 5.0248. However, the reported symbolic regression results focus on the selected feature-subset model, which represents the final model retained by the ScalaTion selection procedure.

Several nonlinear terms were statistically significant, including $\sqrt{\text{age}}$, cement², slag², $\sqrt{\text{water}}$, and water².

6.1.3 Airfoil

The Airfoil symbolic regression model included transformed versions and interactions among frequency, angle of attack, chord length, free-stream velocity, and displacement thickness. Candidate terms included logarithmic, square-root, squared, and multi-variable interaction terms.

Table 17: Selected significant coefficients from the ScalaTion symbolic regression model for Airfoil.

Term	Estimate	Std. Error	<i>t</i>	<i>p</i>
Displacement thickness $\times$ chord length $\times$ frequency	0.7788	0.1640	4.75	< 0.001
Chord length $\times$ frequency	-0.0127	0.0016	-7.81	< 0.001
Attack angle ²	0.1499	0.0302	4.96	< 0.001
Velocity $\times$ chord length $\times$ frequency	0.000171	0.000027	6.34	< 0.001
Custom transformed term	-2.2738	0.2685	-8.47	< 0.001
Chord length $\times$ attack angle $\times$ frequency	-0.00159	0.00020	-8.09	< 0.001
Displacement thickness $\times$ velocity $\times$ attack angle	-0.6896	0.2485	-2.78	0.0055
Displacement thickness $\times$ frequency	-0.1888	0.0456	-4.14	< 0.001
Displacement thickness $\times$ attack angle $\times$ frequency	0.00714	0.00192	3.71	< 0.001
Velocity $\times$ chord length $\times$ attack angle	0.1275	0.0352	3.62	< 0.001
Attack angle ^{1/2}	-24.9059	7.6452	-3.26	0.0011
Frequency	0.00577	0.00142	4.07	< 0.001
Velocity $\times$ chord length	-0.7778	0.1716	-4.53	< 0.001
Velocity ^{1/2}	-3.5519	0.8407	-4.22	< 0.001
Attack angle	280.2162	124.8255	2.24	0.0248
Displacement thickness ^{1/2}	35.0415	10.3336	3.39	< 0.001
Displacement thickness $\times$ velocity	18.7152	5.5923	3.35	< 0.001
Velocity $\times$ attack angle	-0.01508	0.00450	-3.35	< 0.001
Displacement thickness $\times$ velocity $\times$ frequency	-0.000908	0.000452	-2.01	0.0445
Displacement thickness $\times$ chord length	-6170.3137	2631.6606	-2.34	0.0190
$\log(1 + \text{frequency})$	8.6614	1.8632	4.65	< 0.001
$\sqrt{\text{frequency}}$	-0.9257	0.2086	-4.44	< 0.001

The final model achieved an R^2 of 0.6826 and an adjusted R^2 of 0.6321. The RMSE was 3.9647 dB and the MAE was 2.9289 dB. The residual standard error was approximately 4.2752 and the overall F -statistic was 13.5314.

Although the Airfoil model explained less variation than the Auto MPG and Concrete ScalaTion models, it still captured substantial nonlinear structure in the response.

Many of the selected transformed features were statistically significant, indicating that the Airfoil response contains substantial nonlinear and interaction effects. However, several transformed terms had extremely large VIF values, with some approaching infinity. This indicates severe multicollinearity among the generated symbolic features. Therefore, the individual coefficient magnitudes should not be interpreted independently; the model is better viewed as a predictive combination of nonlinear basis functions.

6.2 PySR

PySR was also used to perform symbolic regression on all three datasets. Each dataset was divided into training and testing sets using an 80/20 split with `random_state=42`. PySR searches for mathematical expressions through an evolutionary procedure and evaluates candidate expressions based on predictive loss and expression complexity.

The operator set included the binary operators

$$+, -, \times, /$$

and nonlinear unary transformations including square root and squared terms. For the Concrete and Airfoil datasets, logarithmic transformations were also included.

6.2.1 Auto MPG

The Auto MPG dataset contained 392 observations and seven predictors: displacement, cylinders, horsepower, weight, acceleration, model year, and origin. The data were divided into 313 training observations and 79 testing observations.

PySR was run for 150 iterations with the operators $+$, $-$, $*$, and $/$, together with the `square` and `sqrt` unary operators. Expression complexity was limited using `maxsize=30`, and model selection was performed using the `best` criterion.

The selected equation was

$$\widehat{\text{MPG}} = \frac{2149.9915(\text{model_year} - 46.281776)}{\text{weight}}. \quad (1)$$

This expression indicates that predicted fuel economy increases with model year and decreases with vehicle weight. Despite the availability of seven predictors, PySR selected a compact equation involving only model year and weight.

On the held-out test set, the model achieved

$$\text{MSE} = 7.5596, \quad \text{RMSE} = 2.7495,$$

$$\text{MAE} = 2.0550, \quad R^2 = 0.8519.$$

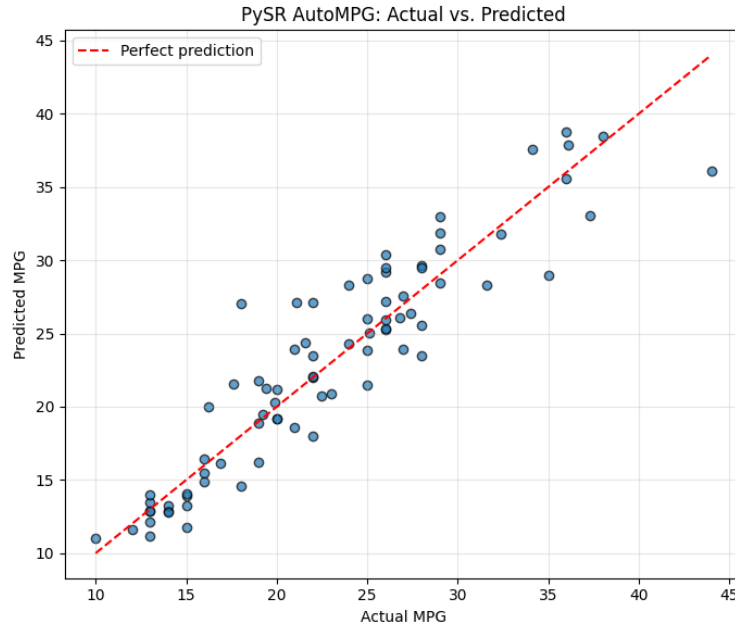


Figure 25: Actual versus predicted MPG values for the PySR symbolic regression model. The dashed diagonal line represents perfect prediction.

The predicted MPG values generally follow the perfect-prediction line, indicating that the compact two-variable equation captures much of the variation in MPG.

6.2.2 Concrete

The Concrete dataset contained 1030 observations and eight predictors: cement, blast furnace slag, fly ash, water, superplasticizer, coarse aggregate, fine aggregate, and age. The dataset was divided into 824 training observations and 206 testing observations.

PySR was run for 300 iterations with arithmetic operators and the nonlinear operators `square`, `sqrt`, and `log`. The final equation selected using the lowest training loss was

$$\widehat{\text{Strength}} = -0.200393 \text{ water} + \sqrt{\text{age superplasticizer}^{1/16}} \quad (2)$$

$$+ 1.201766 \sqrt{\sqrt{\text{fly_ash}} + \text{slag} - \frac{2.197127 \text{ slag}}{\text{age}}} \quad (3)$$

$$+ 6.627109 \log(\text{age}) \quad (4)$$

$$+ 1.201766 \log^2(\text{cement}) \log^2\left(\log\left(\sqrt{\text{cement}}\right)\right). \quad (5)$$

The expression captures several nonlinear relationships. Water contributes negatively, while age enters through both square-root and logarithmic terms. The model also includes nonlinear contributions from slag, fly ash, superplasticizer, and cement.

The held-out test performance was

$$\text{MSE} = 42.3301, \quad \text{RMSE} = 6.5062,$$

$$\text{MAE} = 5.3603, \quad R^2 = 0.8357.$$

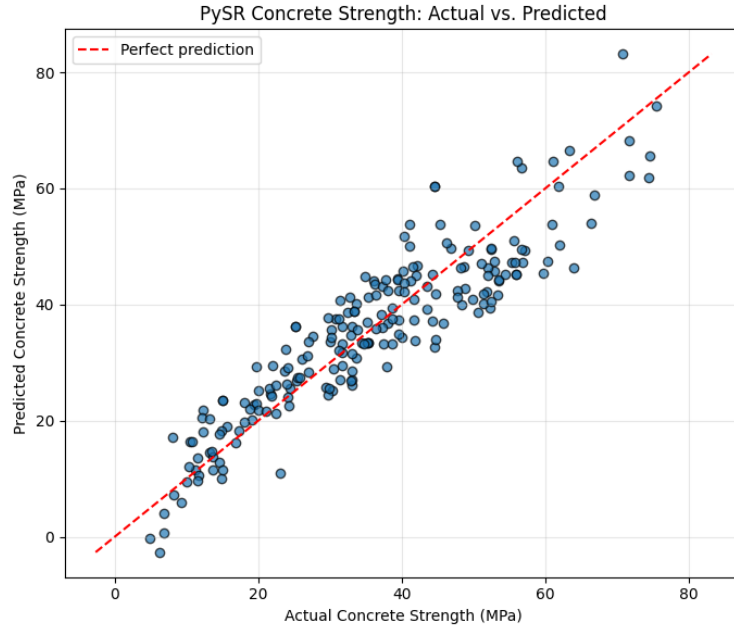


Figure 26: Actual versus predicted concrete compressive strength from the PySR symbolic regression model.

The model explains approximately 83.6% of the variation in compressive strength on the test data, indicating that the nonlinear expression captures important relationships that are not represented by a simple linear model.

6.2.3 Airfoil

The Airfoil Self-Noise dataset contained 1503 observations with five predictors: frequency, angle of attack, chord length, free-stream velocity, and displacement thickness. The data were divided into 1202 training observations and 301 testing observations.

PySR was run for 500 iterations using arithmetic operators together with square, square-root, and logarithmic transformations. The selected symbolic expression was

$$\widehat{\text{SPL}} = - \frac{36.989017 f}{f + (-195.3614 d^2 v + v)^2 + 128.70335 + \frac{2.3002949}{cd}} \quad (6)$$

$$+ 136.76428 - \frac{530.60527}{cf \log(195.3614 d/c)}, \quad (7)$$

where f represents frequency, v represents velocity, c represents chord length, and d represents displacement thickness.

The model obtained the following test performance:

$$\text{MSE} = 9.2244, \quad \text{RMSE} = 3.0372,$$

$$\text{MAE} = 2.3529, \quad R^2 = 0.8159.$$

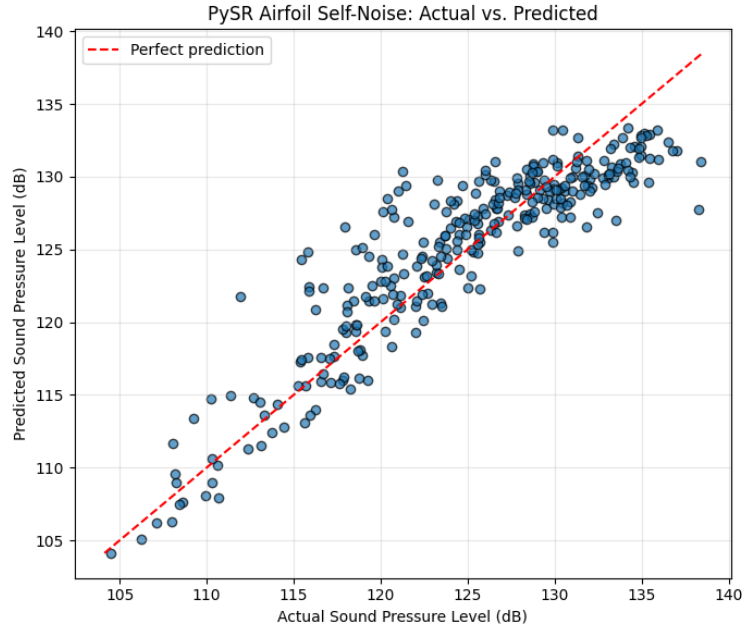


Figure 27: Actual versus predicted sound pressure levels from the PySR symbolic regression model.

The symbolic equation combines frequency, velocity, chord length, and displacement thickness through nonlinear ratios and logarithmic terms. The model explains approximately 81.6% of the variation in sound pressure level on the test set.

6.3 Comparison of ScalaTion and PySR

Table 18: Comparison of ScalaTion and PySR symbolic regression performance on each tool’s own held-out test set.

Dataset	Method	R^2	RMSE
Auto MPG	ScalaTion	0.8827	2.6101
	PySR	0.8519	2.7495
Concrete	ScalaTion	0.8180	6.9646
	PySR	0.8357	6.5062
Airfoil	ScalaTion	0.6826	3.9647
	PySR	0.8159	3.0372

The two symbolic regression implementations produced different equation structures because they use different search procedures. ScalaTion constructs a large transformed feature matrix and performs feature selection, whereas PySR directly searches over mathematical expressions. The two tools were also evaluated on different 80/20 splits (Scala’s random shuffle for ScalaTion and scikit-learn’s `train_test_split` for PySR), so small differences in R^2 and RMSE partly reflect which observations fell in each test set.

For Auto MPG, both methods produced strong predictive performance. ScalaTion reported the larger R^2 , while PySR produced a substantially more compact equation involving only model year and weight. For Concrete, PySR achieved a higher R^2 and lower RMSE than the ScalaTion selected model. The largest difference occurred for the Airfoil dataset, where PySR achieved an R^2 of 0.8159 compared with 0.6826 for ScalaTion.

Overall, the results demonstrate one of the major advantages of symbolic regression: the ability to discover nonlinear functional relationships while still producing explicit mathematical expressions that can be inspected and interpreted.